Band-1 Computational Experiment: Andrica's Conjecture — Finite √-Gap Verification

Human + ZoraASI full-dash (executor)

2026-09-11

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Honesty stamp. This note sieves primes to a box-feasible X and checks

$$\Delta\sqrt{p}_n := \sqrt{p_{n+1}} - \sqrt{p_n} < 1$$

for every consecutive pair with $p_{n+1} \le X$. It records the maximum and its argmax. It does **not** prove Andrica's conjecture; it does **not** elevate 2025–2026 preprint claims of a proof; it does **not** claim Clay / ToE / GATE-04 unlock. **Finite verification only; status = OPEN.**

Abstract

Statement (Andrica, 1988). For all $n \ge 1$, $\sqrt{p_{n+1}} - \sqrt{p_n} < 1$, equivalently $g_n := p_{n+1} - p_n < 2\sqrt{p_n} + 1$.

Box result. With $X = 10^8$, among $\pi(X) = 5{,}761{,}455$ primes (5,761,454 consecutive gaps), **every** $\Delta\sqrt{p}_n < 1$. The observed maximum is

$$\max = \sqrt{11} - \sqrt{7} \approx 0.670873479290809$$

attained uniquely (in this range) at the pair $(p_n, p_{n+1}) = (7, 11)$ (1-based index $n = 4$). Wall $\approx 1.72\,\text{s}$.

Literature. Conjecture **OPEN**. Empirical verification via maximal-gap tables extends to $\sim 2 \times 10^{19}$ (Wikipedia; Cook 2026-04-08; Visser, *Mathematics* **6**(12):289, 2018, via the 80th maximal gap). Box X is an independent finite re-check, not a record.

Problem statement

Check Andrica's inequality on all consecutive primes $\le X$. Report:

1. X, $\pi(X)$, number of gaps;
2. $\max_n \Delta\sqrt{p}_n$ and argmax pair;
3. count of violations $\Delta\sqrt{p}_n \ge 1$ (expect 0);
4. decade checkpoints; unit tests;
5. literature ledger with status **OPEN**.

Literature ledger (not re-proved on box)

Item	Status	Note
Andrica conjecture	**OPEN**	Unproved for all n
Empirical verification (lit)	known far	via maximal gaps $\sim 2 \times 10^{19}$
High-water mark	classical	$A_4 = \sqrt{11} - \sqrt{7} \approx 0.670873\ldots$
Gap bound $g_n < p_n^{0.525}$	theorem (context)	implies Andrica only up to $\sim 1.1 \times 10^{12}$; not full proof

Item	Status	Note
2025–2026 "proof" preprints (Zenodo/SSRN)	**NOT elevated**	no standing peer acceptance assumed here

Primary lit pointers:

- Wikipedia, *Andrica's conjecture* (empirical section; Ghory / maximal-gap extension to 2×10^{19}).
- John D. Cook, *Andrica's conjecture on the gap between root primes* (2026-04-08): verified to 2×10^{19}.
- M. Visser, *Variants on Andrica's Conjecture…*, Mathematics **6**(12):289 (2018): holds through $p^*_{80} > 1.836 \times 10^{19}$.

Method (box)

1. **Sieve of Eratosthenes** (numpy boolean) on $[0, X]$, $X = 10^8$.
2. Extract primes; compute $\Delta\sqrt{p}_n$ in float64.
3. Track max, argmax, top-5, and any ≥ 1 violations.
4. Unit tests: first five primes; $\pi(10^3) = 168$; $\pi(10^6) = 78498$; max to 1000 equals $\sqrt{11} - \sqrt{7}$ at $(7, 11)$.
5. **No claim** that finite success implies the conjecture.

Float64 is adequate: at $X = 10^8$, $\sqrt{p}$ spacing and gaps keep $\Delta\sqrt{p}$ far from cancellation issues relative to the threshold 1.

Results (numbers!)

Unit tests

`PASS`: primes $[2, 3, 5, 7, 11]$; $\pi(10^3) = 168$; $\pi(10^6) = 78498$; max $\Delta\sqrt{p}$ on primes ≤ 1000 is $\sqrt{11} - \sqrt{7}$ at $(7, 11)$; all < 1 on that range.

Primary run ($X = 10^8$)

Quantity	Value
X	100,000,000
$\pi(X)$	5,761,455
Gaps checked	5,761,454
$\max \Delta\sqrt{p}$	$\approx 0.670873479290809$
Argmax pair	(7, 11) (index $n = 4$)
Argmax gap g_n	4
Violations ≥ 1	0
All strictly < 1	**True**
Wall	≈ 1.72 s (sieve + scan)

Decade checkpoints (max $\Delta\sqrt{p}$ among primes $\leq 10^k$)

For every decade $10^k \leq X$, the maximum remains $\sqrt{11} - \sqrt{7}$ at $(7, 11)$ (the classical high-water mark), and all gaps satisfy Andrica.

Top-5 largest $\Delta\sqrt{p}$ in range

See `results.json scan.top5`. The unique maximum is $(7, 11)$; subsequent large values are early primes (consistent with asymptotic decay of typical $\Delta\sqrt{p} \sim g_n/(2\sqrt{p_n})$).

Honesty / non-claims

- **Does claim:** finite verification that $\Delta\sqrt{p}_n < 1$ for all consecutive primes with $p_{n+1} \leq 10^8$; max and argmax recorded.
- **Does NOT claim:** Andrica proved; conjecture closed; elevation of preprint "proofs"; Clay / ToE / GATE-04 unlock.
- **Status: OPEN**.

One-liner: Finite verification only; Andrica remains OPEN — NOT proved.

Reproducibility

```
cd experiments/band1/andrica
python3 andrica_verify.py
# → results.json, literature_ledger.json, SUMMARY.txt
```

Artifacts: `experiments/band1/andrica/`; log `experiments/logs/2026-09-11_andrica.log`; this paper under `mission-control/papers/`.

References

1. D. Andrica, *Note on a conjecture in prime number theory*, Studia Univ. Babeș-Bolyai Math. **33** (1988), 3–7 (historical attribution).
2. Wikipedia, *Andrica's conjecture* (empirical verification section).
3. J.D. Cook, *Andrica's conjecture on the gap between root primes*, 2026-04-08.
4. M. Visser, *Variants on Andrica's Conjecture with and without the Riemann Hypothesis*, Mathematics **6**(12):289 (2018), DOI 10.3390/math6120289.